



Bio-Regenerative Life Support Systems in Space Exploration: Principles, Technologies and the Role of Microorganisms in Sustainable Nutrient Recycling for Long- Term Space Mission: A Literature Review

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Abstract

As humanity pursues long-duration spaceflight, sustainable life support systems are essential to sustain crew survival. In recent times, Bio-regenerative Life Support Systems (BLSS) have emerged due to its potential of integrating biological, physicochemical, and engineering technologies to ensure the recycle of air, water, nutrients, and waste within closed-loop environments and also supporting the production of sustainable food. This study synthesizes evidence on the BLSS for long-term space exploration. This review provides a comprehensive synthesis of recent evidence regarding the principles, technologies, and current innovations in BLSS, with a focus on microbial-driven nutrient recycling. A structured search was conducted on Web of Science, PubMed, and Google Scholar using key search terms. Literature relevant to the review were synthesized qualitatively and data on principles, photobioreactor technologies, artificial intelligence technology for environmental monitoring, synthetic biology, microbial protein production, and waste management and recycle strategies. Key advances in utilizing bacteria, microalgae, fungi, and cyanobacteria facilitate efficient nutrient recycling, respiratory oxygen carbon dioxide interaction, water purification, and resource recovery. Operational challenges include ecological instability, microbial contamination, microgravity interruption, nutritional quality of microbial biomass and long-term reliability of the system. The integration of microbial food systems into BLSS promotes resource efficiency, environmental resilience, and sustainability. These systems also reduce the dependence on earth resupply. However, further

research is required to optimize cultivation systems, enhance long-term operational reliability, and validate these technologies in actual space environments. Although this review highlights the potential of BLSS to support human space exploration, several critical knowledge gaps remain. Thus, this review provides insights on advancement in BLSS technologies and identifies future directions to improve BLSS sustainable human habitation in space environment.

Keywords: Bio Regenerative Life Support System, Life Support Systems, Environmental Control and Life Support Systems, Microbial Food System.

Introduction

In the twenty-first century, human's quest for prolonged space exploration has been driven by the need to establish sustained human habitation beyond the low earth orbit. Through the Artemis Program, the National Aeronautics and Space Administration (NASA) and the European Space Agency (ESA) amongst other international partners have led strategic programs for longer space missions to the moon and future exploration to Mars (NASA, 2020). This ambitious vision requires the establishment of life support infrastructures that are autonomous and resilient to extraterrestrial environments, ensuring long-term survival and independence on the low earth orbit. In order to address the threat to achieving man's quest for space habitation, environmental control and life support systems (ECLSS) efficiently manages air revitalization, recovery of water, waste management and the production of food while minimizing Astronauts' dependence on food and water resupply (Shevtsov et al., 2022). NASA's International Space Station (ISS) depends greatly on the regenerative ECLSS, recycling 90% of water from astronaut's sweat and urine (Gaskil, 2023). However, the physical and chemical based Environmental Control and Life Support System (ECLSS) on the ISS is unable to produce food on-site (Cheng et al., 2025).

Several decades of space research have advanced both theoretical and technological foundations leading the development of Bio-regenerative Life Support Systems (BLSS). This is a close-looped

artificial ecosystems that leverages on biological processes to sustain human survival in resource-limited environment (Sustainability Directory, 2024). It is a system that replicates earth's natural ecological cycle, using plants, algae, microbes and even animals to convert metabolic waste to usable resources including food, water and oxygen (Vunnam, 2024). BLSS also allows for biological and physico-chemical sub-system coupling to ensure material closure and enhance the presence of living systems. Closed-loop systems seek to recycle resources internally, which lessens the operational load for long-duration missions beyond Earth orbit, in contrast to open-loop systems, where consumables must be continuously provided from Earth (*Read "Advanced Technology for Human Support in Space" at NAP.Edu, n.d.*). As a result, scientific explorations have focused on technologies that can sustain the life of astronauts without dependence on the earth's provisions.

Microorganism can potentially be an important part of the closed loop system giving its high efficiency in recycling material and its contribution to environmental homeostasis. Microorganisms are crucial because of their role in converting organic waste into valuable resources that can utilized by crops as nutrients (Matassa et al., 2016). One of the most promising applications of microbes is the synthesis of microbial protein, also known as single-cell protein (SCP), which may provide astronauts with a sustainable food source (Matassa et al., 2016).

Microbial protein can be produced from carbon dioxide (CO₂), human waste, and organic residue. This helps recycle nutrients and reduces waste accumulation especially in a BLSS (Matassa et al., 2016). Thus, the integration of microorganisms into BLSS can help achieve environmental balance that recycles water, regenerates oxygen, and provides the nutrients that are valuable in sustaining human space exploration.

Microalgae are emerging photosynthetic microorganisms that have potential benefits in the closed loop system. These microorganisms have highly efficient photosynthetic capabilities, grow rapidly, and plays multifunctional roles in the production of biomass, oxygen generation, removal of CO₂, and the recycling of nutrient (Yang et al., n.d.). Microalgae such as *Chlorella vulgaris* and *Spirulina* species have been studied and identified as microalgae that can efficiently convert CO₂ to oxygen through the biological process of photosynthesis (Helisch et al., 2020). Through this process, biomass that contain proteins, vitamins and essential amino acids are produced. Another experimental study reveals that microalgae cultivation systems have the ability to maintain oxygen production the generate biomass in closed-loop systems which can potentially sustain air revitalization, supplementation of food and the recycling of waste products making it valuable for long-term space mission (Dai et al., 2026a; Yang et al., n.d.).

Research Problem

Space research has evolved from ecological and closed-system studies to explore potential and complexities of attaining stable bio-geochemical cycles in confined environments, to the optimization of plant productivity and efficiency of bioreactor in space environment, the incorporation of artificial intelligence for monitoring this system, and promoting resilience against extreme environments. Thus, BLSS provides a platform for more advanced research and technological innovations to promote sustainable humans' space habitation. Thus, a robust review of current research on BLSS provides an opportunity to adequately understand advancement in technologies while identifying key areas for development and improvement of the system to ensure sustainable space habitation.

This review narratively explores the BLSS as it focuses on its principles, technologies and potential developments to promote space research and long-term human habitation of space. This will be achieved by;

- Synthesizing existing research and information about the design and functioning principles (biological, chemical and physical) of BLSS
- Evaluating the recent advancements in technology and innovations that have enhanced system efficiency and sustainability
- Reviewing existing knowledge of closed-loop life support systems and microbial protein production.
- Reviewing the biological and biochemical attributes of microalgae relevant to BLSS.
- Reviewing photo-bioreactor technologies and cultivation methods suitable for space environments.
- Identifying challenges and recommending areas for development of the system towards achieving the goal of sustainable space habitation

Methods

Design

This study adopts a structured systematic review to comprehensively synthesize all information on BLSS for space exploration. This review was conducted between 2nd of March till 29th June 2026.

Search Strategy and Search Term

Targeted search was conducted on web of science, PubMed and Google Scholar to identify relevant literature to be included in the review. Key search term include “bio-regenerative life support systems”, ‘OR’ “closed ecological system”, ‘OR’ “closed-loop system”, ‘OR’ “environmental control and life support systems”, ‘AND’ “principles” ‘OR’ “technologies” ‘OR’ “microorganism”, OR’ “microalgae”, “bacteria”, OR’ “space photo-bioreactor”.

Inclusion Criteria

All publications including peer-reviewed articles, websites, seminal publications, documents, books and other relevant publications with information on bio-regenerative life support systems and conducted in the past 1- years (2016-2026) were eligible for inclusion. Other grey articles on the BLSS were however, excluded from the review.

Data Extraction and Synthesis

All important information that are relevant to study objectives was retrieved and included in the review. Thus, the author’s name, publication year, study objectives, study design, key findings, and limitations of the study was extracted and arranged in a tabular format. This information was qualitatively synthesized to identify the principles, biological and technological aspects, performance and operational strategies of the BLSS, potential innovations and current challenges with the system. Information on microorganism and microalgae function in recycling nutrients, technologies for producing proteins using microbes and integration of microbial bioreactors in a closed-loop system and identify future directions and research to improve the system and promote human long-term space exploration was synthesized.

Results and Discussion

Design and Functioning Principles of Bio-regenerative Life Support Systems

BLSS defines a highly integrated ecological-engineering systems that has been designed to support long-term space habitation through the process of repeated recycling of biological resources and the management of closed-loop environment. BLSS is an ecosystem that is either semi or fully enclosed. Through the utilization of biological, engineering and physicochemical resources, oxygen, water, nutrients, and food are recycled while waste are managed efficiently and the stability of the atmosphere is ensured (De Micco et al., 2023; H. Liu et al., 2021; Nwachi et al., 2025; Rossi, 2020; Verbeelen et al., 2024). Also, biological resources like plants, microalgae, cyanobacteria, fungi, bacteria, and engineered microbial consortia are utilized to establish regenerative ecological cycles that are similar to terrestrial ecologies (Sachdeva et al., 2021; Verseux et al., 2016).

The biological regenerative process of nutrient recycling is the foundation of BLSS. Biologically, BLSS adopts the principle of trophic interactions between producers, consumers, and decomposers. Producers include photosynthetic organisms (microalgae, cyanobacteria, and higher plants) that absorb respiratory by-products of astronauts, while ensuring that light energy is converted to oxygen and edible biomass (De Pascale et al., 2021; Verseux et al., 2016). On the other hand, decomposers comprise of

microorganisms that process and convert Astronauts' metabolic wastes as well as food residues, microbial by-products, and mineralize organic matter into reusable inorganic micronutrients including nitrogen, phosphorus, sulfur, and potassium. All these processes collectively function to ensure the sustainability of ecological nutrient loops especially for long-term stability (Fahrión et al., 2021).

Other studies have also shown that the architecture of modern BLSS adopts the interconnectedness of biological and engineering subsystems and follows the principle of compartmentalization to minimize ecological disturbance and improve operational stability (Häder et al., 2018; Paradiso & De Pascale, 2021; Rossi, 2020). These subsystems are specialized nodules that include anaerobic digestion units, algal photobioreactors (PBR), crop cultivation chambers, microbial bioprocessing units, nitrification reactors, and water purification systems that are interconnected to prevent cascading failures and improve efficiency of ecological processes (Häder et al., 2018; Paradiso & De Pascale, 2021; Rossi, 2020). The Micro-Ecological Life Support System Alternative (MELiSSA) model is widely studied as a framework which integrates these compartmentalized nodules, and food production for the recovery of material and regulation of the environment (Paradiso & De Pascale, 2021).

Chemically, carbon, nitrogen, phosphorus, sulfur, and trace mineral cycling in BLSS depend on tightly regulated nutrient transformation pathways. Through microbial mineralization and physicochemical processing, metabolic waste from human, food residues, and microbial degradation are converted into reusable inorganic nutrients (Rossi, 2020; Venkata Mohan et al., 2020). In order to minimize nutrient depletion and maintain biomass productivity in enclosed ecosystems, Nitrogen and phosphorus recovery pathways are initiated (Płaczek et al., 2017). On the other hand, water recovery systems are indispensable components of ecological system as it recycles wastewater by the processes of biological and physicochemical purification while ensuring that nutrients are recovered and reused (Ting et al., 2017; Zamfirescu et al., 2024).

Under microgravity conditions, alteration in the interaction of gas-liquid, transportation of nutrients, transfer of heat, suspension of biomass and oxygen diffusion occurs within the cultivation systems (Hogle et al., 2023). Regardless of these challenges, BLSS must maintain stable environmental conditions physically and in engineering. Thus, environmental stability includes the optimum maintenance of pressure, temperature, humidity, airflow, illumination, radiation shielding, gas exchange, and fluid dynamics (De Pascale et al., 2021). In the absence of buoyancy-driven transport, modern BLSS engineering integrates capillary-driven circulation systems, low-shear circulation mechanisms, membrane-assisted gas exchange technologies, mechanical agitation systems, and automated environmental regulation platforms as a compensatory mechanism (Schäfer, 2025).

BLSS functions as a complex but yet adaptive ecosystem that requires the integration of various disciplines such as ecology, microbiology, materials science, biotechnology, systems biology, aerospace engineering, environmental engineering and artificial intelligence. Although there has been significant progress in ensuring BLSS, there are still challenges in complete ecological closure and long-term operability which is a huge limitation to sustainable extraterrestrial habitation.

Recent Technological Advancements and Innovations Enhancing Bio Regenerative Life Support System's Efficiency and Sustainability

In recent times, substantial technological advancement in BLSS research has emerged. This includes areas of PBR engineering, synthetic biology, systems automation, ecological optimization, environmental monitoring, and hybrid life-support integration all of which have significantly enhanced the feasibility, efficiency, and sustainability of BLSS and the prospects for lunar, Martian, and deep-space missions.

The evolution of PBR technologies for algal and cyanobacterial cultivation has been identified as one of the most important technological advancements in BLSS research. This technology adopts a highly controlled closed photobioreactor rather than the traditional open cultivation system. Also, the technology ensures the optimization of photosynthesis while minimizing loss of water, the risk of contamination, and operational instability (Hoque et al., 2025; Keller et al., 2023). Some of the most efficient systems for extraterrestrial applications include flat-panel, tubular, membrane-assisted, bubble-column, rotating, and modular reactor configurations as the geometry of the reactor enhances the penetration of light, efficiency in the transfer of gas, the distribution of nutrients, and biomass productivity (Cortês et al., 2020). Specifically, flat-panel systems are suitable for confined habitats giving them the ability to optimize surface-area-to-volume ratios while the system improves photon utilization and minimizes spatial requirements (Hogle et al., 2023).

Giving the limited weight requirement and the instability of microgravity especially during long-term space missions, space-oriented cultivation systems utilize systems and technologies incorporated to withstand fluid instability due to microgravity, increased exposures to radiation and challenges with operational control. This includes, lightweight materials, modular construction, adaptive environmental control, membrane-assisted gas exchange, and energy efficient circulation technologies and systems (Schäfer, 2025). Also, biofilm-based and immobilized cultivation systems have been identified as valuable alternatives to suspended cultures. This is because the dependence on gravitational mixing with immobilization which makes it easier to harvest biomass, improves operational stability, and reduces the risk of contamination (Lasseur & Mergeay, 2021; Mazzolini et al., 2026). Evidence from a previous study on *Chlorella vulgaris* revealed that calcium alginate bead immobilization significantly improved the efficiency of nutrient uptake and retention of biomass in a closed-loop system (Mazzolini et al., 2026).

Advances in environmental automation, artificial intelligence (AI), and predictive computational modeling in BLSS have also been identified. Incorporation of these technologies provides monitoring systems that are sophisticated with sensors that can regulate atmospheric composition, dissolved oxygen, nutrient concentrations, pH, biomass density, illumination intensity, and water quality independently and in real time. As a result, the operational reliability is enhanced while the workload on astronaut is greatly minimized especially during long-term space missions (I. Ahmad et al., 2021). Predictive control systems and machine-learning algorithms infused with AI technology when integrated into cultivation systems can enhance the flow of nutrients, prevent unstable ecologies, and detect any malfunctioning of subsystem leading to prompt intervention and prevention of catastrophic events. This model also has the capacity to detect metabolic imbalance, contamination, and precariousness of subsystems.

Lighting technologies are another advancement that have been reported in literature. Under extraterrestrial conditions, LED systems with enhanced wavelengths and energy-efficiency have been shown to improve photosynthetic productivity while minimizing the consumption of electrical energy (Paradiso & De Pascale, 2021). Scientific evidence shows that the combination of red and blue spectral in microalgal cultivation systems can optimize the accumulation of biomass and photosynthetic efficiency (Li et al., 2020). Another review supports the suggestion that LED lighting (red and blue) technologies can promote sustainable cultivation as tunable LED luminaires trigger photosynthesis which in turn influences plant shape, improve nutrient and plant biomass while reducing electrical energy wastage, the buildup of heat and minimize workload (Rahman et al., 2021).

Another innovative area of the BLSS is synthetic biology and metabolic engineering. It has been shown in a study that microorganisms and algal strains that have been genetically optimized have enhanced

photosynthetic efficiency and the assimilation of nutrients (Fahrion et al., 2021; Hoque et al., 2025). Also, genetically enhanced microorganisms and algae are shown to have high radiation resistance, stress tolerance, carbon fixation capabilities as well as improved protein productivity (Fahrion et al., 2021; Hoque et al., 2025). Giving these potentials, biologically engineered systems can also prospectively contribute to innovations in pharmaceuticals and the production of high antioxidants, vitamins, and therapeutic biomolecules to meet astronaut health needs especially during long term space missions. Microbiome engineering that incorporates a community of various interactive microorganisms have also been found to enhance resilience, the efficiency of nutrient recycling, metabolic symbiosis, and environmental consistency and resilience unlike monoculture systems (Schultz et al., 2026).

The integration of in-situ resource utilization (ISRU) like Martian atmospheric CO₂, extraterrestrial water resources, and regolith-derived minerals as substrates for biological cultivation systems into the BLSS have been largely explored (Glukhikh, 2022; Verseux et al., 2016). The integration of physicochemical and biological modules has also been explored as a hybrid-life support system with potential of ensuring environmental stability by the physicochemical and sustainable regenerative sustainability for the biological system.

These systems reveal technological innovation and advancements in BLSS research. However, these innovations remain at pilot stages. Hence, there is need for long-term testing exploration of extraterrestrial conditions, uncertainties that relate to energy optimization, ecological stability, control of contamination, resilience against radiation, and operational stability.

Closed-Loop Life Support Systems and the Production of Microbial Proteins

Due to the dynamic nature of biological systems and its high sensitivity to environmental changes, include imbalance in microorganism, a significant limitation is the ability of BLSS achieving complete ecological closure. However, BLSS research and innovation remains the most significant gateway for human exploration of extraterrestrial habitat as it can significantly minimize human dependence on resupplies from earth and ensure valuable recycling of biological resources (De Micco et al., 2023; H. Liu et al., 2021). Through ecological interaction and a series of metabolic pathways, water, gases, nutrients and organic waste are recycled into reusable biological substrates (Rossi, 2020; Sachdeva et al., 2021).

Microbial protein production is one evolving BLSS innovation with potential for meeting the nutritional needs of astronauts during deep-space missions. Unlike in extraterrestrial habitations, the adoption of terrestrial agriculture would depend on high demand for water, land, energy consumption with relatively low productivity which makes it unsuitable for the confined extraterrestrial habitat conditions (Poughon et al., 2020). In contrast, microbial biomass production systems including algae, fungi, yeast, bacteria cyanobacteria, offer substantially higher resource efficiency, productivity, and compatibility as wastes are also recycled (Heinicke & Arnhof, 2021; Sachdeva et al., 2021).

Microorganisms produce single-cell proteins (SCP) that have amino acid profiles that are similar to conventional protein sources and do not require much environmental resources to grow (Heinicke & Arnhof, 2021). Some of the microorganisms like *Chlorella vulgaris*, *Arthrospira platensis*, and *Spirulina platensis* have been identified as valuable microbial food sources (Barone et al., 2023). This is because these microbes are enriched with high protein, amino acid profiles, antioxidants, vitamins, and polyunsaturated fatty acids (Barone et al., 2023). Additionally, studies have reported microalgal species that were cultivated under optimized environmental cultivations had protein concentrations of about 40% to 70% dry biomass (Ali et al., 2025; Mosibo et al., 2024; Pulz & Gross, 2004; Saeed et al., 2025; Song et al., 2025). Additionally, studies found that microbial cultivation systems can be a valuable

contribution to atmospheric revitalization, the recycling of nutrients, the rectification of wastewater, capture of carbon, and stabilization of the ecology within integrated BLSS architectures (Nyugen et al., 2022; Rossi, 2020; Verseux et al., 2016). As a result, nutrient-rich cultivation media are generated from the mineralization of organic waste streams which can establish nutrient cycle and improve sustainability (Sabri, 2018). The nitrogen and phosphorus derived from waste and recovered through microbial degradation can be harnessed for algal cultivation, promoting the circulation of materials and minimizing the loss of resources. A study reported that algal and microbial biomass contain key nutrients that can ameliorate oxidative stress, radiation-induced cellular damage and immune dysfunction due to spaceflight (Pulz & Gross, 2004). These nutrients include essential amino acids, antioxidants, vitamins, carotenoids, chlorophyll pigments, omega-3 fatty acids, and bioactive compounds. Another study found that extraterrestrial ecosystems can benefit from engineered microbial community's ability to detoxify its habitat, environmental stabilization, and biomedical resilience within extraterrestrial ecosystems (Adhikary et al., 2026).

Several advancements in research have shown the potential of incorporation of microbial proteins into the BLSS. However, some areas including long-term nutritional adequacy, digestibility of biomass, sensory evaluation of microbial proteins, potential of elevation of nucleic acid concentrations, accumulation of toxins, metabolic indices, and psychological acceptance continue remains a significant limitation to the integration of microbial proteins into the BLSS to meet astronaut's dietary needs (Poughon et al., 2020). There is also a tendency of large-scale SCP systems experiencing heightened risk of contamination, while the high demand for energy and instability of operations in a closed-loop conditions remains a challenge. Thus, it is important that further innovations and development of downstream technologies are explored. These include cell disruption, refining of biomass, fermentation optimization, metabolic engineering, and extraction of protein as this will improve bioavailability and safety of nutrient and long-term dietary acceptability of astronauts.

Biological and Biochemical Characteristics of Microalgae in Bio-regenerative Life Support Systems

Microalgae is an important component of the BLSS as it has great potentials due to its physiological versatility, photosynthetic efficiency and biochemical qualities. In addition to microalgae's capability of generating higher amounts of oxygen and edible biomass per unit cultivation compared to terrestrial crops, it requires lesser volume of water, growth time and ecosystem which allows for an improved efficiency of resources in a confined extraterrestrial ecosystem (Keller et al., 2023; Ravindran et al., 2016). (Life, 2023; Wheeler, 2020). *Chlorella vulgaris*, *Spirulina platensis*, *Arthrospira*, *Euglena gracilis*, and various cyanobacteria have been extensively researched because of their potential for rapid biomass accumulation, enhanced photosynthetic efficiency, its ability to adapt to controlled environments, and superior nutritional qualities (Barone et al., 2023; Verseux et al., 2016). In particular, *Chlorella vulgaris* has been found to have good tolerance to higher CO₂ concentrations as observed in spacecrafts and its consistent productivity under confined cultivation conditions (Niederwieser et al., 2018).

Microalgae also possess good biochemical properties including its high nutritional and biomedical value. Studies have shown that biomass generated from microalgae high proteins, essential amino acids, vitamins, antioxidants, minerals, carotenoids, chlorophyll pigments, and polyunsaturated fatty acids content (Pulz & Gross, 2004). The nutrients are most especially important for the maintenance of astronauts' nutrition and physiological health maintenance (Pulz & Gross, 2004; Sonali et al., 2025). Studies have also shown that bioactive compounds in microalgae have substantial antioxidant, anti-inflammatory, immunomodulatory and cyto-protective properties which makes them relevant for

mitigating radiation-induced oxidative stress and immune dysregulation during long-term space travel (Dai et al., 2026b; Eze et al., 2023; Finamore et al., 2017; Hoque et al., 2025; Wu et al., 2016).

Microalgae are highly resilient extreme environmental conditions and metabolically flexible (A. Ahmad & Ashraf, 2024; Praveen et al., 2024). Many microalgal species have the capacity to tolerate increased concentration of CO₂, while withstanding unstable nutrient availability, atmospheric pressures, limited water availability, desiccation, and exposure to high radiation exposure all of which are poor conditions of extraterrestrial habitations (Hogle et al., 2023). Many of these strains are considered extremophiles because of their ability to cope with chronic exposure to radiation and other environmental stressors, making them suitable for inclusion into the BLSS especially as man continue to explore long-term space exploration (Montenegro-Herrera & Maroneze, 2026).

Additionally, microalgae has been researched for their strong metabolic adaptability (Sajadian et al., 2018). It has been reported that many microalgal species have the capacity to grow under autotrophic, heterotrophic, or mixotrophic conditions depending on environmental parameters and resource availability (Bastos & Altenhofen Da Silva, 2026). This capability enhances their ability to be compatible with various configurations of BLSS. As reported in other studies microalgae efficiently assimilate nitrogen and phosphorus that are produced from wastewater streams, which makes them efficiently recycle nutrients and reduce high dependence on fertilizers produced externally (Akuku & Satognon, 2025; Delgadillo-Mirquez et al., 2016; Fernandes et al., 2017; Popa et al., 2025).

Beyond the production of food and oxygen the strains that are rich in lipids can support the generation of biofuels. Considering many species are rich in antioxidant and carotenoids, they can be beneficial to providing physiologic protection considering the potential of the space adverse environmental in inducing oxidative damage (Guo et al., 2017). Genetically engineered microalgae have the capacity for innovations in pharmaceuticals, vitamins, nutraceuticals, and therapeutic production of biomolecules even within extraterrestrial ecosystems (Grama et al., 2022; Shelake et al., 2025).

Within algal cultivation systems, microbial interactions have been identified. Specifically, studies suggest that through a symbiotic approach, non-axenic cultivation strategies can potentially improve the recycling of nutrients, metabolic synergy, and their resilience extreme ecologies (Akuku & Satognon, 2025; Benaisaa et al., 2025; Xing et al., 2025). It is however important to note that if uncontrolled, there is potential for microbial contamination which can simultaneously result in ecological instability including biofilm fouling and pathogen proliferation. BLSS studies are taking into consideration the adoption of ecological engineering to regulate microbial consortia and promote a controlled microbial community (De Micco et al., 2023; Verbeelen et al., 2021). For example, The top-down approach applies physical and chemical pressures including the alteration of pH, temperature cycling and dilution to explore the dynamics of microbial communities and this approach utilizes natural or mixed ecological communities (Lyu et al., 2024). The bottom-up approach utilizes synthetic microbial consortium to assemble microbial strains and leverage on their interactions through a process of cross feeding. This in turn, ensures the stability of microbial communities and prevent microbial dominance (Saccardo et al., 2022).

Some of the biological limitations identified include photoinhibition due to dense cultures, poor light penetration, and self-shading (Saccardo et al., 2022). There are also issues regarding the accumulation of oxygen, metabolic instability, and the formation of biofilm all of which are important consideration in BLSS and prolonged space habitation (Fahrion et al., 2021). Furthermore, there are scientific concerns about the long-term impact of microalgal exposure to extreme space conditions on metabolism, gene expression, and productivity (Dai et al., 2026c). However, existing research works have focused on the selection of microalgal strains, the evolution in their adaptation, metabolomics, and

the potential of genetic engineering in improving productivity, resilience and nutritional benefits under extreme environmental conditions.

Photobioreactor Technologies and Cultivation Methods Suitable for Space Environments

Studies in the review identify photobioreactor technology as an important component to ensure BLSS feasibility. As a result, space-based photobioreactors must be efficient and ensure photosynthetic productivity, contamination control, and ecological balance while minimizing energy consumption, conserve water and reliably operating in space environment (Keller et al., 2023). According to studies reviewed, a promising quality of closed-loop photobioreactor systems was the ability to improve environmental regulation, resource recovery and atmospheric control while minimizing the risk of contamination (Hoque et al., 2025; Sangregorio-Soto et al., 2026; Thajuddin et al., 2026).

The configuration of some photobioreactor have been examined for its application in extraterrestrial environments. These systems include flat-panel reactors, bubble-column reactors, airlift systems, tubular systems, membrane photobioreactors, rotating bioreactors, and modular cultivation systems (Carone et al., 2024; Cortesão et al., 2020; Eilertsen et al., 2022; Huang et al., 2016; Kura et al., 2025; X. Liu, 2026; Peng et al., 2016a, 2016b; Poughon et al., 2020; Schäfer, 2025). Flat-panel bioreactors have been reported to be the most efficient reactor for extraterrestrial conditions because of its high surface area to volume ratio which allows for the efficient use of light, exchange of gas and control of temperature. For example, the photobioreactor at the life support rack (PBR@LSR) on the International Space Stations (ISS) adopts a flat-panel airlift photobioreactor that is compatible with microgravity (Detrell et al., 2019). The PBR@LSR uses a hybrid life support system and integrate both biological and physiochemical processes to cultivate *Chlorella vulgaris* (Detrell et al., 2019). The cultivation *Chlorella vulgaris* cultivated was used in the conversion of CO₂ to oxygen and this was successful accounting for the first space-based hybrid algae life support system and the most valuable strategy for the future Lunar and Martian Space mission (Detrell et al., 2019).

Also, the bubble-column bioreactor is a cylindrical reactor that allows gas to be introduced at the base through a sparger which allows for the mixing and delivery of CO₂ without mechanical interruption (Fahrion et al., 2021; O'Higgins et al., 2024). Because of its simple design, low energy requirement, sterilization ease and its salability, bubble-column bioreactors have been widely adopted for the cultivation of microalgae and its potential for integration into the BLSS (Fahrion et al., 2021; O'Higgins et al., 2024). It has been shown that these reactors are efficient in gas-liquid transfer and capacity for high biomass productivity which makes them an option for integration into the future Lunar and Martian mission (Fahrion et al., 2021; O'Higgins et al., 2024). One limitation is its bubble movement under microgravity which could negatively affect as-liquid separation (Fahrion et al., 2021; O'Higgins et al., 2024). However, a modified configuration that is assisted by a membrane system can alleviate these challenges.

Tubular bioreactors are transparent enclosed microalgal cultivation tubes that are packed either horizontally, vertically, helically or serpentine (Chandola et al., 2026; Fahrion et al., 2021). These reactors have high surface area to volume ratio which allows for the capture of light, superior biomass production and mitigation of contamination unlike open cultivation systems (Chandola et al., 2026; Fahrion et al., 2021). For future mission such as Lunar and Martian, this system is mostly important as it has potential to efficiently generate oxygen, remove CO₂ and aid in the production of biomass (Chandola et al., 2026; Fahrion et al., 2021). While ensuring a controlled environment, tubular bioreactor can achieve natural or artificial illumination by incorporating airlift or pump circulation systems (Chandola et al., 2026; Fahrion et al., 2021). However, some limitations such as the

accumulation of oxygen, poor gas-fluid exchange, biofouling are to be considered to improve the system and its feasibility under microgravity condition.

In space environments, microgravity significantly impacts fluid circulation, nutrient transport, gas-liquid separation, and the suspension of biomass. Consequently, membrane-assisted gas exchange technologies, capillary-based fluid systems, low-shear circulation mechanisms, and mechanical agitation systems are capable of addressing the limitation of poor buoyancy. Membrane photobioreactor adopts membrane-based separation and gas-exchange technologies for microbial cultivation making them ideal for BLSS (Fahrion et al., 2021; Poughon et al., 2020). Adopting a buoyancy bubble separation technology, semi-permeable membranes are used for CO₂ and oxygen transfers while ensuring high biomass productivity (Fahrion et al., 2021; Poughon et al., 2020). Experiments on the ISS have cultivated *Limnospira indica*, using membrane-separated bioreactors and found continuous cyanobacterial growth and the production of oxygen under space conditions (Poughon et al., 2020). This makes membrane-based bioreactors viable for future BLSS research for long-term space exploration where CO₂ recycling and biomass production is important.

Exploring immobilized and biofilm-based cultivation systems have been identified as important alternatives to suspended cultures. All these technologies have helped in reducing cultivation system's dependence on gravitational suspension, while ensuring ease of biomass harvesting, retention, and promote resilience and stability to microgravity conditions (Lasseur & Mergeay, 2021; Mazzolini et al., 2026). Microalgae immobilization including calcium alginate beads functions by alleviating complexities of fluid management and enhancing biomass recovery under controlled conditions.

Another photobioreactor technology is the rotating photobioreactor which is a cultivation system that uses the principle of rotation to promote mixing and the distribution of light and production of biomass in suspension while limiting biofouling (Kretschmer et al., 2022; Wood et al., 2022). As a result, this technology does not rely on gravity making it an important technology for BLSS in space environment (Kretschmer et al., 2022; Wood et al., 2022). The rotating technology allows microbial culture vessels to be exposed to a light and dark cycle that alternates. As a result, the technology promotes high biomass productivity, while conserving energy and limiting mechanical disruptions (Kretschmer et al., 2022; Wood et al., 2022). Similarly, the rotating wall vessels have been adopted as a tool to simulate microgravity and for ground-based space research. Thus, rotating photobioreactor technology is valuable for future Lar and Martian BLSS especially considering the need for sustainable generation of oxygen, high biomass production and utilization of resources.

Although there are a lot of innovation on the integration of BLSS in long term space exploration, some challenges have been identified that continue to negatively impact photobioreactors integration in space. Some factors such as light inhibition, poor/uneven absorption of light, self-shading, the accumulation of oxygen, challenges with regulating temperature and biofouling affect photosynthetic productivity (Hoque et al., 2025; Paradiso & De Pascale, 2021). Also, long periods of exposure to cosmic radiation could threaten biological productivity and hamper the integrity of bioreactor materials. There is therefore the need for more advancement in technology to improve the design of reactors, ensure optimum environmental regulation, enhance materials engineering, and fluid management technologies to achieve future deployment of BLSS in space.

Challenges and Future Innovations towards Sustaining Long-term Space Habitation

This review has identified significant scientific and technological innovations in BLSS. However, studies in this review reports several limitations in the biological, engineering, ecological, and operational quality of these systems thus hindering fully attainment of autonomy and sustainability in

space environment. One major concern is the sustainability of ecological stability within the closed-loop system. Giving the need for biological systems to be dynamic and sensitive to the uncertainties of extreme environments and other challenges including microbial imbalance, contamination, fluctuation in nutrients and unstable atmospheric events (Ciurans Molist, 2023). Studies have shown that changes in environmental conditions can potentially affect microbial architecture and affect ecological networks and potentially catastrophic system failure making it difficult to achieve complete ecological closure (Hernandez et al., 2021; König et al., 2018; Shaw et al., 2020; Strona & Lafferty, 2016).

Although microgravity has been reported to have beneficial impact on plant cultivation systems, the extreme space ecological condition remains a significant obstacle to productivity and performance. This is because microgravity alters the cultivation system's process of fluid distribution, gas exchange, biofilm formation, nutrient transportation, biomass suspension, cellular physiology and the diffusion of oxygen (Hogle et al., 2023). While other technologies such as membrane-based gas exchange, immobilized cultivation systems amongst others are promising, there is need for examining of long-term operationality in microgravity environment.

Space radiation is also an important challenge especially for biological organisms during long-term space missions. Although some microorganisms are extremophiles and are resilient, long-term exposures to radiation can undermine photosynthesis, microbial community, photobioreactor materials while inducing DNA damage. Also, the long-term effect of exposure to extremophiles has also been significantly understudied.

Photobioreactors often require high energy for its operation including illumination, thermal control, environmental regulation, atmospheric circulation, water purification, and biomass production (Keller et al., 2023; Paradiso & De Pascale, 2021). Therefore, future bioreactors must be energy efficient in ensuring that the architecture, integration, waste recovery, environmental technologies, and resource allocation systems are maintained and sustainable.

BLSSs are highly susceptible microbial contamination, biofouling, proliferation of pathogens leading to an unstable ecology. As a result, these systems must be efficient in sustainably ensuring sterilization and microbial control in order to achieve optimum operationality and while promoting the health of astronauts. Also, it is important that ecological engineering should explored to maintain microbial consortia as some microorganisms are beneficial to promote resilience and recycling of nutrients. There are also concerns about nutritional adequacy and acceptability of foods that are biologically derived. While microalgal biomass are nutrient dense and have valuable bioactive compounds, there are concerns about how long-term dependence on limited dietary diversity can negatively impact astronauts in terms of their physiological and psychological wellbeing including nutritional balance. Therefore, there is need for strategies to improve nutritional optimization, sensory acceptability and dietary diversities of astronauts.

There are also issues with how scalable and engineering procedures of the BLSS are ensured under a highly controlled laboratory conditions and how this can replicate the complexities of extraterrestrial environment (Kolodziejczyk & Summerer, 2016; Smith et al., 2025; Zamfirescu et al., 2024). Also, the transitioning from prototypes to integration into space habitat must require advancement in technologies such as modularity, automation, integration of systems and tolerance/resilience against faults during long-term experiments for validation. Also, it has been shown that there are psychological and social impact of BLSS. All biological forms including cultivation systems, plant-based ecologies can impact the health of astronauts as it promotes psychological and emotional stability, visual stimulation and ecological cohesion during long-term space mission where there is confinement and isolation (Paradiso & De Pascale, 2021).

Studies included in this review have found BLSS technologies to be the most viable strategy for human habitation in space. Also, microorganisms, advanced photobioreactor, and hybrid regenerative systems are advantageous given their multi-functionality, adaptability, sustainability and regenerative potential. However, there is need for more innovations in areas of synthetic biology, systems ecology, materials science, ecological engineering, microbiome engineering, artificial intelligence, and in-situ resource utilization in order to achieve sustainability in extraterrestrial habitats.

Conclusion

One of the most significant space technologies in recent time is BLSS as it supports sustainability and independence in man's quest to inhabit the space environment. This review reports the BLSS operates on the principles of closed-loop and space engineering to synergistically recycle biological components like air, water and nutrients that would support human's prolonged survival especially the production of highly nutritious space food in space environment. Thus, BLSS offers prospects for autonomy and resource efficiency and the numerous studies conducted using analogue terrestrial habitation, space-based innovation and controlled ecological system have reported great potential and the possibility of integrating these approaches into space support architectures while taking into consideration waste reduction, resource management in extraterrestrial environments,

Another key finding in this review is the importance of microorganism in contributing to nutrient production, efficiency and sustainability and its potential of recycling and supporting the functionality of life within the BLSS. Biogeochemical processes such as decomposition of organic waste, mineralization of nutrients, recycling of carbon and mineralization of nutrients are highly supported by microbial communities and this ensures the availability and sustainability of biological resources that ensures human and plant survival. However, some operational challenges include the instability of microorganisms, the inability to ensure resilience and complete sustainability of systems, other extreme environmental stressors and incorporation into large-scale. It is important that further research address these limitations and encourage interdisciplinary approach that will integrate biotechnology, microbiology, space sciences and engineering in achieving an ultimate goal of resilience, efficiency, sustainability and robustness. Amongst the numerous technologies that have been developed and implemented in space, the deployment of a robust microbial nutrient recycling system is key for ensuring self-sustainability and supporting future space explorative quest such as Lunar and Martian space missions.

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Conflict of Interest

The authors declare no conflict of interest.

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